Future Architecture – Zecpath AI

High-Level Architecture

```
                Frontend Dashboard
                        |
                        ▼
                API Gateway Layer
                        |
```

——————————————————————————————————— AI Intelligence Layer ———————————————————————————————————

• Resume Parser

• ATS Engine

• Screening AI

• HR Interview AI

• Technical Interview AI

• Machine Test AI

• Behavioral AI

• Video Analysis Engine

• Emotion Detection Engine

• Decision AI

• AI Coaching Engine

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```
          Analytics & Learning Platform
```

• Candidate Database

• Report Generator

• Dashboard Analytics

• Model Registry

• Feedback Learning Engine

• Data Lake

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```
            Cloud Infrastructure
```

• Docker

• Kubernetes

• Redis

• PostgreSQL

• Object Storage

• Monitoring

Future Microservices

- Resume Service
- ATS Service
- Interview Service
- Technical Evaluation Service
- Machine Test Service
- Behavioral Analysis Service
- Integrity Detection Service
- Decision Service
- Report Service
- Analytics Service

AI Learning Loop

Candidate Interview

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Recruiter Feedback

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Performance Database

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Model Retraining

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Updated AI Models

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Improved Future Decisions

Scalability Strategy

- Microservice architecture
- Horizontal scaling
- GPU inference servers
- Distributed processing
- Cloud-native deployment
- Continuous model updates

Long-Term Vision

Build a fully autonomous AI recruitment platform capable of conducting interviews, evaluating candidates, generating reports, learning from recruiter feedback, and continuously improving hiring quality.